

Start Early & Compounding

BEGINNER

LESSON EXPLAINER

■ What Is Compounding?

Compounding just means your money makes money — and then that money starts making money too. Imagine you invest \$100 and it grows by 10% in a year. You now have \$110. The next year, that 10% growth applies to the full \$110, not just your original \$100 — so you earn \$11 instead of \$10. It sounds small, but stretched over decades, that snowball effect becomes enormous.

Think of a snowball rolling down a long hill. At the top, it barely grows — there isn't much snow to stick to it yet. But the further it rolls, the bigger it gets, and the bigger it gets, the faster it picks up even more snow. Money invested early is that snowball right at the top of a very long hill.

■ Real Example: Sarah vs. Mike

Sarah starts investing \$200 a month at age 25. Mike waits ten years and starts investing \$400 a month — twice as much — at age 35. Both keep going until they retire at 65, and both earn a typical long-run average return of 7% a year.

	Sarah — starts at 25	Mike — starts at 35
Monthly investment	\$200	\$400
Years invested	40 years	30 years
Total contributed	\$96,000	\$144,000
Estimated value at 65	\$524,963	\$487,988

Even though Mike put in **50% more of his own money**, Sarah still ends up ahead — purely because her money had ten extra years to compound. That's the entire lesson in one example.

■ The Power of Reinvesting the Gains

A one-time \$10,000 investment growing at 7% a year, compared two ways: reinvesting each year's growth (compound) versus taking the same 7% only on your original \$10,000 every year (simple). Same rate, same starting amount — the only difference is whether the gains stay invested.

Year	Compound (reinvested)	Simple (not reinvested)	Extra from reinvesting
Year 1	\$10,700	\$10,700	+\$0
Year 2	\$11,449	\$11,400	+\$49

Year	Compound (reinvested)	Simple (not reinvested)	Extra from reinvesting
Year 3	\$12,250	\$12,100	+\$150
Year 4	\$13,108	\$12,800	+\$308
Year 5	\$14,026	\$13,500	+\$526
Year 6	\$15,007	\$14,200	+\$807
Year 7	\$16,058	\$14,900	+\$1,158
Year 8	\$17,182	\$15,600	+\$1,582
Year 9	\$18,385	\$16,300	+\$2,085
Year 10	\$19,672	\$17,000	+\$2,672
Year 11	\$21,049	\$17,700	+\$3,349
Year 12	\$22,522	\$18,400	+\$4,122
Year 13	\$24,098	\$19,100	+\$4,998
Year 14	\$25,785	\$19,800	+\$5,985
Year 15	\$27,590	\$20,500	+\$7,090
Year 16	\$29,522	\$21,200	+\$8,322
Year 17	\$31,588	\$21,900	+\$9,688
Year 18	\$33,799	\$22,600	+\$11,199
Year 19	\$36,165	\$23,300	+\$12,865
Year 20	\$38,697	\$24,000	+\$14,697
Year 21	\$41,406	\$24,700	+\$16,706
Year 22	\$44,304	\$25,400	+\$18,904
Year 23	\$47,405	\$26,100	+\$21,305
Year 24	\$50,724	\$26,800	+\$23,924
Year 25	\$54,274	\$27,500	+\$26,774
Year 26	\$58,074	\$28,200	+\$29,874
Year 27	\$62,139	\$28,900	+\$33,239
Year 28	\$66,488	\$29,600	+\$36,888

Year	Compound (reinvested)	Simple (not reinvested)	Extra from reinvesting
Year 29	\$71,143	\$30,300	+\$40,843
Year 30	\$76,123	\$31,000	+\$45,123

Based on a \$10,000 starting amount at a 7% annual return — the same illustrative figure used above, not a guarantee of future performance.

Worth knowing: the 7% figure above is a simplified, illustrative long-run average — real markets go up and down year to year, and past performance never guarantees future returns. The point isn't the exact number; it's the principle that time is one of the most powerful tools you have.

■ Quick Check: 5 Questions

Answer all five before turning the page to check yourself against the answer key.

1. What does "compounding" mean in investing?

- A. Your returns start earning their own returns over time
- B. Only your original deposits ever grow
- C. Your bank charges you extra interest on loans
- D. You must manually reinvest your dividends every single day

2. Why can starting 10 years earlier have such a big impact, even with smaller monthly contributions?

- A. Because markets always rise in an investor's first 10 years
- B. Because the extra decade gives compounding more time to work
- C. Because younger investors are charged lower fees by law
- D. It doesn't really matter — timing has no real effect

3. In the Sarah vs. Mike example, why does Sarah end up with more money despite contributing less overall?

- A. She chose better-performing investments than Mike
- B. Mike invested in a much riskier type of account
- C. Her contributions had 10 extra years to compound
- D. Sarah received a large one-off tax refund

4. Which statement best describes the "cost of waiting" to start investing?

- A. There's no real cost — you can always catch up later by investing more
- B. It only matters if you're investing more than \$1,000 a month
- C. Waiting is actually better because prices always fall over time
- D. Delaying reduces the time compounding has to work, which can be very hard to fully make up

5. What's the single biggest takeaway from this lesson?

- A. You need a large lump sum of cash to start investing
- B. You need to be a market expert before compounding helps you
- C. Time in the market is one of the most powerful tools for building wealth
- D. Compounding only works if you invest more than \$1,000 a month

■ Quiz Answer Key

- 1 **A** — Compounding means your returns start earning their own returns over time.
- 2 **B** — An extra decade gives compounding more time to work — that's why starting early matters so much.
- 3 **C** — Sarah ends up ahead because her contributions had 10 extra years to compound, not because of better investment choices.
- 4 **D** — Delaying investing reduces the time compounding has to work, which can be very hard to fully make up later.
- 5 **C** — Main takeaway: time in the market is one of the most powerful tools for building wealth.